

Tu Li

Curriculum Vitae (Last updated: September 5, 2026)

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Education

Ph.D. Candidate, Economics	2022–Present
University of Arizona, Tucson, USA	
<i>Committee:</i> Hidehiko Ichimura (Chair), James L. Powell, Tiemen Woutersen	
M.S. Mathematics	2022
California State University, Long Beach, USA	
B.S. Economics	2019
California State Polytechnic University, Pomona, USA	

Research Interests

Econometric theory; semiparametric estimation and efficiency; moment selection and regularization.

Working Papers

Penalized and Debiased GMM with Potentially Invalid Moment Conditions

Abstract. This paper proposes a penalized debiased generalized method of moments (PDG) procedure for estimation and moment selection when candidate moment conditions may be invalid and the moment functions depend on first-step nonparametric or machine-learning estimators. We assume that a set of valid baseline moments identifies the parameter of interest, while additional candidate moments may improve estimation precision when valid. To account for the plug-in step bias, we construct the orthogonal and cross-fitted sample debiased moments for potentially invalid moments, building on Chernozhukov et al. (2022). We show that the proposed procedure has oracle-like properties: it consistently selects the correct moment conditions, and its asymptotic distribution is the same as that of the infeasible oracle GMM, even when the moments contain estimated first-step components, for a fixed data-generating process. The simulation results show that PDG accurately recovers the invalid set while exhibiting finite-sample shrinkage bias, and that the post-selection PDG, which we recommend for final estimation and inference, has negligible bias and provides substantial variance reductions relative to baseline-only GMM.

Teaching Experience

University of Arizona

Sole Instructor

ECON 453: Data Analytics and Modeling: Quantitative Analysis for Economic Strategy

In person; Fall 2026; Spring 2026

Covers applied statistics and machine learning, including regression, regularization, nonparametric methods, and resampling. Developed original lecture slides and notes to modernize the course.

ECON 330: Macroeconomic and Global Institutions and Policy

Online; Summer 2026; Summer 2025

Examines financial markets and institutions, monetary and fiscal policy, and business decisions.

Teaching Assistant

ECON 407: Economics of Strategy

ECON 200: Basic Economic Issues

ECON 501A: Microeconomic Theory (PhD level)

Presentations

Metric Lunch, University of Arizona

Skills and Languages

Programming: R, Python

Languages: Chinese (native); English (fluent)

Other Work Experience

Tea Barista and Cashier, Half & Half, Los Angeles area, CA

2014–2017

References

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